

Machine Learning I — Hands-On 10 — Worked Solutions

10.1 From transactions to itemsets

In association rule mining, the data are represented as transactions containing sets of items. The exercise uses the following 10 transactions.

Transaction ID	Items bought
T1	Bread, Milk, Butter
T2	Bread, Milk
T3	Bread, Eggs
T4	Milk, Butter
T5	Bread, Milk, Butter, Eggs
T6	Bread, Butter
T7	Milk, Eggs
T8	Bread, Milk, Butter
T9	Bread, Milk, Coffee
T10	Coffee, Eggs

For $N = 10$ transactions,

$$\text{support}(X) = \frac{\#\{t : X \subseteq t\}}{10}.$$

1. Support of one-item itemsets

Bread

Bread appears in T1, T2, T3, T5, T6, T8, and T9: 7 transactions.

$$\text{support}(\{\text{Bread}\}) = \frac{7}{10} = 0.70 = 70\%.$$

Milk

Milk appears in T1, T2, T4, T5, T7, T8, and T9: 7 transactions.

$$\text{support}(\{\text{Milk}\}) = \frac{7}{10} = 70\%.$$

Butter

Butter appears in T1, T4, T5, T6, and T8: 5 transactions.

$$\text{support}(\{\text{Butter}\}) = \frac{5}{10} = 50\%.$$

Eggs

Eggs appears in T3, T5, T7, and T10: 4 transactions.

$$\text{support}(\{\text{Eggs}\}) = \frac{4}{10} = 40\%.$$

Coffee

Coffee appears in T9 and T10: 2 transactions.

$$\text{support}(\{\text{Coffee}\}) = \frac{2}{10} = 20\%.$$

The one-item results are therefore:

Itemset	Count	Support
{Bread}	7	70%
{Milk}	7	70%
{Butter}	5	50%
{Eggs}	4	40%
{Coffee}	2	20%

2. Support of larger itemsets

Bread and Milk

Both items occur together in T1, T2, T5, T8, and T9: 5 transactions.

$$\text{support}(\{\text{Bread, Milk}\}) = \frac{5}{10} = 50\%.$$

Bread and Butter

They occur together in T1, T5, T6, and T8: 4 transactions.

$$\text{support}(\{\text{Bread, Butter}\}) = \frac{4}{10} = 40\%.$$

Milk and Butter

They occur together in T1, T4, T5, and T8: 4 transactions.

$$\text{support}(\{\text{Milk, Butter}\}) = \frac{4}{10} = 40\%.$$

Bread, Milk, and Butter

All three occur together in T1, T5, and T8: 3 transactions.

$$\text{support}(\{\text{Bread, Milk, Butter}\}) = \frac{3}{10} = 30\%.$$

Coffee and Eggs

They occur together only in T10: 1 transaction.

$$\text{support}(\{\text{Coffee, Eggs}\}) = \frac{1}{10} = 10\%.$$

Itemset	Count	Support
{Bread, Milk}	5	50%
{Bread, Butter}	4	40%
{Milk, Butter}	4	40%

Itemset	Count	Support
{Bread, Milk, Butter}	3	30%
{Coffee, Eggs}	1	10%

3. Frequent itemsets for $\text{minsup} = 30\%$

An itemset is frequent when its support is at least 30%.

Therefore, the frequent itemsets among those considered are:

- {Bread};
- {Milk};
- {Butter};
- {Eggs};
- {Bread, Milk};
- {Bread, Butter};
- {Milk, Butter};
- {Bread, Milk, Butter}.

{Coffee} and {Coffee, Eggs} are not frequent because their supports are 20% and 10%, respectively.

4. Why support is not implication strength

Support measures **how often an itemset occurs**. For example, {Bread, Milk} has support 50%, meaning that Bread and Milk appear together in half of the transactions.

However, support alone does not tell us how often Milk appears **given that Bread appears**. The itemset is symmetric: {Bread, Milk} and {Milk, Bread} are the same set. To describe a directional implication, we need confidence.

i ML insight — frequency and direction are different questions

Support is a property of a set of co-occurring items. Confidence is a property of a directional rule. The same joint frequency can therefore produce different rule confidences when the antecedents have different frequencies.

10.2 From itemsets to association rules

For a rule $X \rightarrow Y$,

$$\text{support}(X \rightarrow Y) = \text{support}(X \cup Y),$$

and

$$\text{confidence}(X \rightarrow Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)}.$$

Equivalently, using counts,

$$\text{confidence}(X \rightarrow Y) = \frac{\#(X \cup Y)}{\#(X)}.$$

1. Bread → Milk

The union {Bread, Milk} occurs in 5 transactions, while Bread occurs in 7.

$$\text{support} = \frac{5}{10} = 50\%,$$

$$\text{confidence} = \frac{5}{7} \approx 0.7143 = 71.43\%.$$

2. Milk → Bread

The same union occurs in 5 transactions, and Milk also occurs in 7.

$$\text{support} = 50\%,$$

$$\text{confidence} = \frac{5}{7} \approx 71.43\%.$$

3. Bread → Butter

Bread and Butter occur together in 4 transactions, while Bread occurs in 7.

$$\text{support} = \frac{4}{10} = 40\%,$$

$$\text{confidence} = \frac{4}{7} \approx 57.14\%.$$

4. Butter → Bread

The joint support is still 40%, but Butter occurs in only 5 transactions.

$$\text{confidence} = \frac{4}{5} = 80\%.$$

This pair illustrates why reversing a rule leaves support unchanged but can change confidence.

5. {Bread, Milk} → Butter

The three-item set {Bread, Milk, Butter} occurs in 3 transactions. The antecedent {Bread, Milk} occurs in 5.

$$\text{support} = \frac{3}{10} = 30\%,$$

$$\text{confidence} = \frac{3}{5} = 60\%.$$

6. Coffee → Eggs

Coffee and Eggs occur together only in T10, while Coffee occurs in T9 and T10.

$$\text{support} = \frac{1}{10} = 10\%,$$

$$\text{confidence} = \frac{1}{2} = 50\%.$$

The results are:

Rule	Support	Confidence
{Bread} → {Milk}	50%	71.43%
{Milk} → {Bread}	50%	71.43%
{Bread} → {Butter}	40%	57.14%
{Butter} → {Bread}	40%	80%
{Bread, Milk} → {Butter}	30%	60%
{Coffee} → {Eggs}	10%	50%

7–8. Comparing Bread → Milk and Milk → Bread

The two rules have the same support because both use the same union {Bread, Milk}. In this particular dataset, they also happen to have the same confidence because Bread and Milk each have support 70%.

In general, reversing a rule does **not** preserve confidence. If $\text{support}(X) \neq \text{support}(Y)$, then

$$\frac{\text{support}(X \cup Y)}{\text{support}(X)} \neq \frac{\text{support}(X \cup Y)}{\text{support}(Y)}.$$

10.3 Selecting rules using support and confidence

The thresholds are

$$\text{minsup} = 30\%, \quad \text{minconf} = 70\%.$$

A rule is valid only if it satisfies **both** conditions.

Rule	Support	Confidence	Support ≥30%?	Confidence ≥70%?	Valid?
Bread → Milk	50%	71.43%	Yes	Yes	Yes
Milk → Bread	50%	71.43%	Yes	Yes	Yes
Bread → Butter	40%	57.14%	Yes	No	No
Butter → Bread	40%	80%	Yes	Yes	Yes
Bread, Milk → Butter	30%	60%	Yes	No	No
Coffee → Eggs	10%	50%	No	No	No

Therefore, with $\text{minsup} = 30\%$ and $\text{minconf} = 70\%$, the accepted rules are:

- Bread → Milk;
- Milk → Bread;
- Butter → Bread.

If the minimum confidence is increased to 90%, none of these candidate rules remains valid because none reaches 90% confidence.

When the minimum confidence threshold becomes stricter, fewer rules are accepted. The procedure becomes more selective, retaining only rules where the consequent appears very reliably when the antecedent appears. This can remove weak rules, but it can also remove moderately reliable relationships that may still be useful.

10.4 Why confidence can be misleading

Confidence can be high simply because the consequent is common. Lift corrects for the consequent's marginal frequency:

$$\text{lift}(X \rightarrow Y) = \frac{\text{confidence}(X \rightarrow Y)}{\text{support}(Y)}.$$

- lift > 1: positive association;
- lift = 1: approximately independent;
- lift < 1: negative association.

1. Bread → Milk

$$\text{lift} = \frac{0.7143}{0.70} \approx 1.02.$$

The rule is close to independence.

2. Milk → Bread

$$\text{lift} = \frac{0.7143}{0.70} \approx 1.02.$$

Again, the association is close to independence.

3. Coffee → Eggs

$$\text{lift} = \frac{0.50}{0.40} = 1.25.$$

This is a positive association, but its support is only 10%.

4. Eggs → Coffee

The confidence is

$$\text{confidence}(\{\text{Eggs}\} \rightarrow \{\text{Coffee}\}) = \frac{1/10}{4/10} = \frac{1}{4} = 0.25.$$

Therefore,

$$\text{lift} = \frac{0.25}{0.20} = 1.25.$$

The lift is the same as in the reverse direction because lift measures departure from independence symmetrically through the joint occurrence, even though confidence is directional.

The summary is:

Rule	Support	Confidence	Lift	Interpretation
Bread → Milk	50%	71.43%	1.02	approximately independent
Milk → Bread	50%	71.43%	1.02	approximately independent
Coffee → Eggs	10%	50%	1.25	positive association, low support

Rule	Support	Confidence	Lift	Interpretation
Eggs → Coffee	10%	25%	1.25	positive association, low support and confidence

5–6. Interpreting “interesting” rules

Bread → Milk and Milk → Bread have relatively high support and confidence, but their lift is close to 1. Much of their apparent reliability comes from Bread and Milk already being common.

Coffee → Eggs has the largest lift among the listed rules, so it shows the strongest departure from independence. However, the rule is supported by only one joint transaction, so that estimate is based on very little evidence.

i ML insight — interestingness is objective-dependent

There is no universally most interesting rule. If the objective is **coverage**, support matters. If it is **directional reliability**, confidence matters. If it is **departure from independence**, lift matters. A high-lift rule with very low support should still be interpreted cautiously because it is estimated from few observations.